NATIONAL ENGINEERING COLLEGE, K.R. NAGAR, KOVILPATTI – 628 503

(An Autonomous Institution, Affiliated to Anna University – Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING &

DEPARTMENT OF INFORMATION TECHNOLOGY

23CS11E / 23IT11E - WEB FRAMEWORKS USING PYTHON

Odd Semester - 2025-2026

Preparatory Test – 02 Question bank

CO3: Design and develop UI for dynamic web applications using Flask. (CDL2)

Part – A (Two Marks)

1. Distinguish between a dynamic URL and a static URL in Flask.
2. State the function of the url_for() helper function
3. Characterize the role of a secret key in a Flask application.
4. Create a view function show_user that takes a username as a string parameter from a dynamic URL and returns an f-string with the username.
5. A view passes a list of items named shopping_list. Write the Jinja2 code to iterate over and display each item in the list.
6. Write a Flask route that returns the current date and time.
7. Detail the purpose of using a separate templates folder in a Flask project.
8. Mention two features that make Flask a micro web framework.
9. Write a Flask program to create a form with name and email, and handle form submission using POST method.
10. Define the purpose of sessions for managing user state.

Part – B (8 Marks)

1. A small online store needs a Flask application where products are displayed on different pages using clean URLs such as /products/shoes or /products/bags. Formulate the logic for routing in Flask that maps category names from the URL to the correct product page dynamically. Include code snippets to demonstrate @app.route() usage and dynamic parameters.
2. Discuss how template inheritance and static file management can be applied? Take any real time scenario, and evaluate how the approach improves performance and maintainability.
3. Design a Flask web application for an Online Course Registration system where:
    a. A homepage introduces the system.
    b. A list of available courses is displayed.
    c. Students can view details of a course.
    d. Students can register for a course via a simple form.
    e. Registered course details display

4. A college website requires different pages for students, faculty, and administrators. Each type of user should be able to access a unique route that displays information relevant to their role. The task involves creating a Flask application where multiple routes are defined using the @app.route decorator. Each route must return different content (for example, “Student Page,” “Faculty Page,” and “Admin Page”). The specification demands efficient use of URL routing concepts, including dynamic routes where parameters such as user ID or role can be passed directly in the URL. The code must ensure a modular organization of routes for clarity and future scalability.
5. An online registration form for a workshop needs to capture participant details like name, email, and selected sessions. The solution must use standard HTML forms integrated with Flask. The form should include validation to ensure that email addresses are in the correct format and that required fields are not left blank. The submitted data must be processed by Flask and displayed back to the user as a confirmation page
6. In higher education institutions, a single course is often handled by multiple faculty members. Students may wish to choose the faculty of their preference based on teaching style, schedule compatibility, or mentorship. However, each faculty member can accommodate only a limited number of students, and once the capacity is reached, no further registrations should be allowed. Currently, this allocation process is done manually, leading to confusion, over-allotment, and delays. There is a need for a web-based automated system that allows students to register their choice of faculty in real time while ensuring that faculty seat availability is properly managed.

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